

1 Square Lattice

The tight-binding Hamiltonian for a square lattice is

$$H = -t \sum_{m,n} \left(c_{m+1,n}^\dagger c_{m,n} + c_{m-1,n}^\dagger c_{m,n} + c_{m,n+1}^\dagger c_{m,n} + c_{m,n-1}^\dagger c_{m,n} \right). \quad (1)$$

Now applying the Landau gauge

$$\mathbf{A} = (0, Bx, 0), \quad (2)$$

the hopping along the y direction picks up a Peierls phase. Therefore the Hamiltonian becomes

$$H = -t \sum_{m,n} \left[c_{m+1,n}^\dagger c_{m,n} + c_{m-1,n}^\dagger c_{m,n} + e^{i2\pi\alpha m} c_{m,n+1}^\dagger c_{m,n} + e^{-i2\pi\alpha m} c_{m,n-1}^\dagger c_{m,n} \right], \quad (3)$$

where

$$\alpha = \frac{\Phi}{\Phi_0}. \quad (4)$$

Now define the wavefunction as

$$|\psi\rangle = \sum_{m,n} \psi_{m,n} c_{m,n}^\dagger |0\rangle = \sum_{m,n} \psi_{m,n} |m, n\rangle. \quad (5)$$

Here $\psi_{m,n}$ is a complex number, namely the probability amplitude of the particle at the lattice site (m, n) .

Using the anticommutation relation

$$\{c_{m,n}, c_{m,n}^\dagger\} = 1, \quad (6)$$

one obtains the Harper equation

$$-t [\psi_{m+1,n} + \psi_{m-1,n} + e^{i2\pi\alpha m} \psi_{m,n+1} + e^{-i2\pi\alpha m} \psi_{m,n-1}] = E \psi_{m,n}. \quad (7)$$

Because the system is translationally invariant along the y direction, we use

$$\psi_{m,n} = e^{ik_y n} \phi_m. \quad (8)$$

Substituting this into the Harper equation gives

$$-t [\phi_{m+1} + \phi_{m-1} + 2 \cos(2\pi\alpha m + k_y) \phi_m] = E \phi_m. \quad (9)$$

This is the one-dimensional Harper equation for the square lattice.

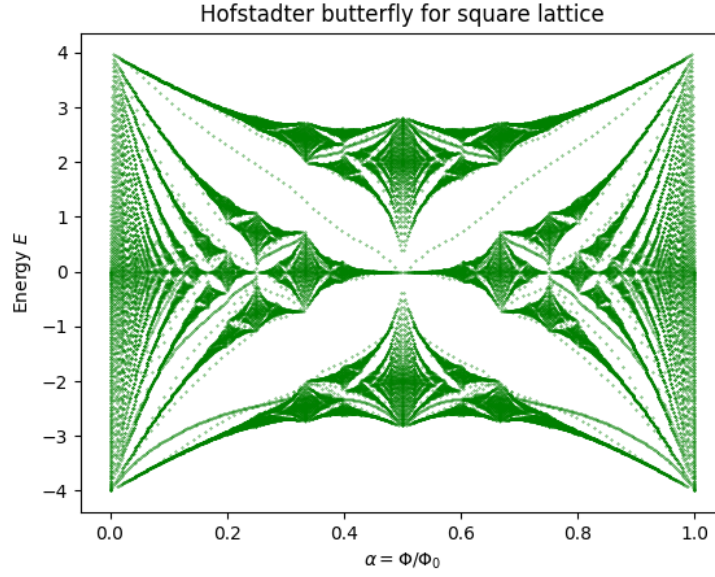


Figure 1: Energy spectrum for the square lattice.

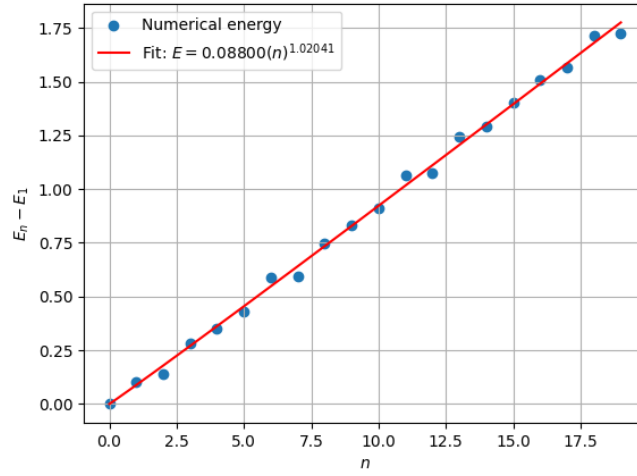


Figure 2: E vs n plot.

2 Hexagonal Lattice with Two Atoms per Unit Cell

For a hexagonal lattice with two atoms in each unit cell, let $c_{m,n}$ denote the annihilation operator on sublattice A and $d_{m,n}$ denote the annihilation operator on sublattice B .

The Hamiltonian is written as

$$\begin{aligned}
 H = & -t' \sum_{m,n} [c_{m,n}^\dagger d_{m,n} + c_{m,n}^\dagger d_{m-1,n} + c_{m,n}^\dagger d_{m,n-1} + \text{h.c.}] \\
 & -t \sum_{m,n} [c_{m+1,n}^\dagger c_{m,n} + c_{m,n+1}^\dagger c_{m,n} + c_{m+1,n-1}^\dagger c_{m,n} \\
 & + d_{m+1,n}^\dagger d_{m,n} + d_{m,n+1}^\dagger d_{m,n} + d_{m+1,n-1}^\dagger d_{m,n} + \text{h.c.}].
 \end{aligned} \tag{10}$$

Now applying the Landau gauge

$$\mathbf{A} = (0, Bx, 0), \tag{11}$$

the Peierls phase is

$$\phi_{ij} = \frac{e}{\hbar} \int_{\mathbf{r}_i}^{\mathbf{r}_j} \mathbf{A} \cdot d\mathbf{l}. \tag{12}$$

For the Landau gauge,

$$\mathbf{A} \cdot d\mathbf{l} = Bx dy. \tag{13}$$

If the lattice site is labelled by (m, n) , with m along the x direction and n along the y direction, then

$$x = ma, \quad y = na. \tag{14}$$

Hence,

$$\phi = \frac{e}{\hbar} Bma^2 \Delta n = 2\pi \alpha m \Delta n, \tag{15}$$

where

$$\alpha = \frac{\Phi}{\Phi_0}. \tag{16}$$

For hopping along the m direction, $\Delta n = 0$, so no Peierls phase appears. For hopping along the n direction, $\Delta n = \pm 1$, and the phase factor becomes

$$e^{\pm i 2 \pi \alpha m}. \quad (17)$$

Therefore the Hamiltonian in the magnetic field becomes

$$\begin{aligned} H = & -t' \sum_{m,n} [c_{m,n}^\dagger d_{m,n} + c_{m,n}^\dagger d_{m-1,n} + e^{i 2 \pi \alpha m} c_{m,n}^\dagger d_{m,n-1} + \text{h.c.}] \\ & -t \sum_{m,n} [c_{m+1,n}^\dagger c_{m,n} + e^{i 2 \pi \alpha m} c_{m,n+1}^\dagger c_{m,n} + e^{-i 2 \pi \alpha (m+1)} c_{m+1,n-1}^\dagger c_{m,n} \\ & + d_{m+1,n}^\dagger d_{m,n} + e^{i 2 \pi \alpha m} d_{m,n+1}^\dagger d_{m,n} \\ & + e^{-i 2 \pi \alpha (m+1)} d_{m+1,n-1}^\dagger d_{m,n} + \text{h.c.}]. \end{aligned} \quad (18)$$

Now define the wavefunction as

$$|\psi\rangle = \sum_{m,n} [A_{m,n} c_{m,n}^\dagger + B_{m,n} d_{m,n}^\dagger] |0\rangle. \quad (19)$$

Using

$$H |\psi\rangle = E |\psi\rangle, \quad (20)$$

we obtain the coupled Harper equations

$$\begin{aligned} E A_{m,n} = & -t' [B_{m,n} + B_{m-1,n} + e^{i 2 \pi \alpha m} B_{m,n-1}] \\ & -t [A_{m+1,n} + A_{m-1,n} + e^{i 2 \pi \alpha m} A_{m,n+1} + e^{-i 2 \pi \alpha m} A_{m,n-1} \\ & + e^{i 2 \pi \alpha (m+1)} A_{m+1,n-1} + e^{-i 2 \pi \alpha m} A_{m-1,n+1}], \end{aligned} \quad (21)$$

and

$$\begin{aligned} E B_{m,n} = & -t' [A_{m,n} + A_{m+1,n} + e^{-i 2 \pi \alpha m} A_{m,n+1}] \\ & -t [B_{m+1,n} + B_{m-1,n} + e^{i 2 \pi \alpha m} B_{m,n+1} + e^{-i 2 \pi \alpha m} B_{m,n-1} \\ & + e^{i 2 \pi \alpha (m+1)} B_{m+1,n-1} + e^{-i 2 \pi \alpha m} B_{m-1,n+1}]. \end{aligned} \quad (22)$$

Now use the plane-wave form along the y direction:

$$A_{m,n} = e^{i k_y n} A_m, \quad B_{m,n} = e^{i k_y n} B_m. \quad (23)$$

For convenience, define

$$\theta_m = k_y + 2\pi\alpha m, \quad \eta_m = 2\pi\alpha m - k_y. \quad (24)$$

Then the Harper equations become

$$EA_m = -t' [(1 + e^{i\eta_m})B_m + B_{m-1}] - t [2 \cos \theta_m A_m + (1 + e^{-i\eta_m})A_{m-1} + (1 + e^{i\eta_{m+1}})A_{m+1}], \quad (25)$$

and

$$EB_m = -t' [(1 + e^{-i\eta_m})A_m + A_{m+1}] - t [2 \cos \theta_m B_m + (1 + e^{-i\eta_m})B_{m-1} + (1 + e^{i\eta_{m+1}})B_{m+1}]. \quad (26)$$

Now write the equations in block-matrix form. Define

$$\Psi_m = \begin{pmatrix} A_m \\ B_m \end{pmatrix}. \quad (27)$$

Then

$$E\Psi_m = h_m\Psi_m + T_m\Psi_{m+1} + T_{m-1}^\dagger\Psi_{m-1}, \quad (28)$$

where

$$h_m = \begin{pmatrix} -2t \cos \theta_m & -t'(1 + e^{i\eta_m}) \\ -t'(1 + e^{-i\eta_m}) & -2t \cos \theta_m \end{pmatrix}, \quad (29)$$

and

$$T_m = \begin{pmatrix} -t(1 + e^{i\eta_{m+1}}) & 0 \\ -t' & -t(1 + e^{i\eta_{m+1}}) \end{pmatrix}. \quad (30)$$

Therefore the full Harper matrix has the block-tridiagonal form

$$H_{\text{Harper}} = \begin{pmatrix} h_0 & T_0 & 0 & 0 & \cdots \\ T_0^\dagger & h_1 & T_1 & 0 & \cdots \\ 0 & T_1^\dagger & h_2 & T_2 & \cdots \\ 0 & 0 & T_2^\dagger & h_3 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}. \quad (31)$$

Solving the eigenvalue equation

$$H_{\text{Harper}}\Psi = E\Psi \quad (32)$$

gives the energy spectrum E as a function of the magnetic flux ratio α .

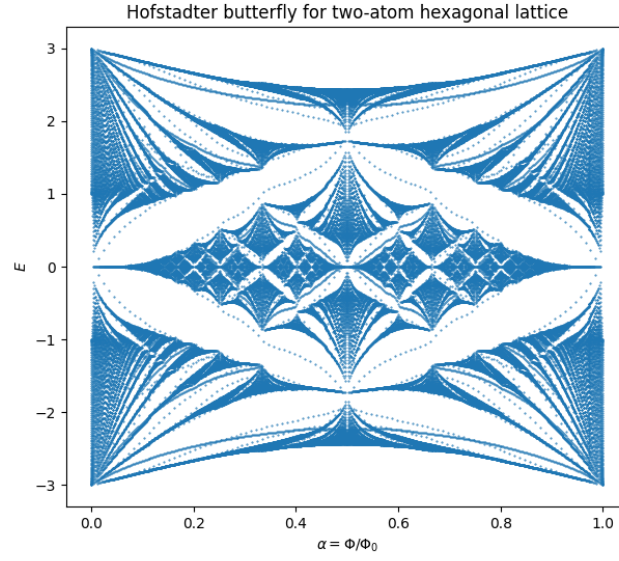


Figure 3: Energy spectrum for the hexagonal lattice.

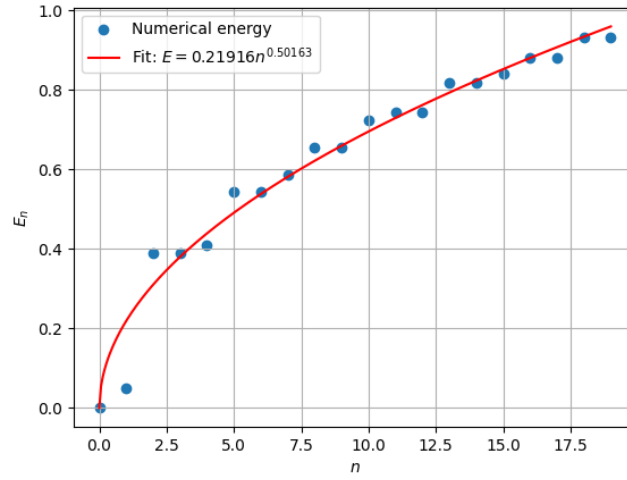


Figure 4: E vs n plot for the hexagonal lattice.